



Gene editing to restore vision

A Gene-Editing Experiment Let These Patients With Vision Loss See Color Again
<https://digit.in/oct21-67>



Brain implant can transmit words

Brain Implant Transmits Full Words from Neural Signals.
<https://digit.in/oct21-68>

advocated the destruction of rats and the elimination of rat infestations in food supplies and grain stores by rat-proofing buildings and also recommended the use of toxins like zinc phosphide, thallium sulfate, arsenic, etc. They also recommended the use of Warfarin, a relatively untested (at the time) anticoagulant rodent poison.

A FLAW IN THE PLAN

Widespread use of Arsenic Trioxide tracking powder in all permanent buildings and farms within control zones was commissioned between 1952 and 1953.

Though effective in the curbing of rat populations, the treatment was haphazard with citizens not being informed about the use of arsenic or being allegedly told that it was only harmful to the rodents. As a result, several inadvertent poisoning incidents occurred among livestock, poultry and pets. Happily, however, the Alberta Agriculture authority sent letters to the citizens in the control zones, warning them against the dangers of the treatment until the eventual discontinuation of the treatment due to high costs and the ever-looming danger of arsenic poisoning.

THE LONG TERM PLAN

The rat control program being used currently can trace its roots to the decisions taken and policies developed back in 1953 to 1959. The appointment of pest control inspectors by municipalities was increased and administered by the local government with support from the provincial government. And rat bait was supplied at no expense to all the municipalities that had an appointed pest control inspector. The recommended rodent poison quickly changed from finely-ground corn with Warfarin to coarsely-rolled oats with Warfarin following field trials that revealed the latter to have better results. Although it was well accepted that rats were unwanted, there was some resistance from the

public especially among some of the indigenous communities, who were hesitant to use rat poison-laced bait as a means of extermination mainly due to the potential danger it posed to livestock, children and pets. Those concerns though were quickly clarified thanks to a particularly brave and committed Alberta Agriculture staff member who ate warfarin-treated rolled oats at a series of meetings with members of Indigenous communities. This alleviated some of the safety concerns among the citizens making them a more proactive part of the rat removal effort.

THE LINE OF DEFENCE

The main areas that bear the brunt of halting the spread of rats even today are the rural municipalities that border the neighbouring province of Saskatchewan in the east. And since those municipalities are responsible for the entire province to be rat-free, the Alberta Agriculture authority agreed to pay 50 percent of all the expenses incurred by the pest control inspectors of each municipality.

One of the main reasons contributing to the rat-free nature of the Alberta province is that rat control is not an optional activity for citizens as well as government officials. Rat control is mandatory and required by law. Failure to comply with rat control guidelines and repeated official warnings from pest control inspectors could result in court action. And it would seem that the possibility of court action and conviction has had the desired result as no more than seven notices by rat control inspectors have been issued in any year since 1950.

THE NEW AGE MEASURES

Since the late 1950's, rat control norms have not changed majorly in Alberta. The share of funding from the provincial government has increased since then, with the government funding 100 percent since 1975. Rat infestations are still eliminated by bait, gas or traps and

the farmers that fall in the control zone still follow the guidelines regarding the elimination of rat food sources and the maintenance of bait stations. With the advances in science, newer anticoagulants are used regularly with Warfarin still being the standard poison.

The number of infestations dropped significantly since 1959. Since then, the occasional rat sightings have been caused mostly by overland freight transports with less than 10 infestations reported in any one year while in the interiors of Alberta have been limited to single rat sightings.

Keeping white rats or rats of any kind has also been made illegal as an escaped pet could multiply and spread through Alberta. Zoos, universities and recognised research or educational institutes, however, are exceptions to the rule.

The autumn of 2014 also saw the introduction of a new phone number that citizens could use to report rat sightings and an email address was set up in 2020 where locals could send pictures and locations of suspected rat infestations. Although hundreds of suspected infestations are reported every year, most end up being misidentified, muskrats, ground squirrels or bushy-tailed woodrats. All reports are still investigated thoroughly though. Albertans in the control zone are still briefed on the identification and prevention of rat infestations although most already know every trick in the book after 70 long years of war on rats.

The last major incident of rats in Alberta occurred back in 2004 when someone released 38 rats in the city of Calgary. The rat control offices though did not have much to do when they arrived as most rats were dead thanks to the efforts of the local residents. A fine of CAD 190,000 (CAD 5000x 38 rats) would have been levied on the buffoon who let the rats loose, had he been caught. As of today, the province of Alberta remains rat free. 